

CHIITECH

AI Business Assistant & Audit Companion for Nigerian SMEs

PRODUCT BLUEPRINT • MVP • UI/UX DESIGN • AUDIT MODULE • PRICING • TECHNICAL ROADMAP

A practical business operating system for small businesses to record sales, control stock, understand profit, stay audit-ready, and use AI to make better decisions.



Recommended standard price: ₦5,000/month

Founding-user price: ₦2,500/month for the first 50-100 businesses

1. Product Vision

Problem: Small businesses often keep sales, expenses, customer debts and stock in notebooks, WhatsApp chats or disconnected tools. Owners may know their cash balance but not their real profit, whether their records would survive scrutiny, or what action to take next.

Solution: ChiiTech combines simple business records with an AI assistant that answers questions using the owner's business data, and an audit layer that keeps every record clean, traceable and ready for a bank, investor, tax authority or external auditor to inspect at any time.

Positioning: “Your AI business manager and built-in auditor” — not simply another POS, and not a bolt-on accounting tool.

Core AI questions

- How is my business doing today?
- Which products make me the most money?
- What should I restock?
- Why did my profit fall?
- Who owes me money?
- Are there any transactions that look unusual or risky?
- Is my business audit-ready right now?
- Create a WhatsApp advert for my new products.
- Give me a simple plan to increase sales this week.

2. MVP Feature Set

Module	What it does	Priority
Dashboard	Sales, expenses, profit, stock alerts and quick actions	Must have
Products	Products, prices, quantities, categories and stock levels	Must have
Sales	Record sales and automatically update stock	Must have
Expenses	Record operating costs	Must have
Customers	Profiles, purchase history and debts	Must have
Reports	Daily/weekly/monthly sales and profit	Must have
Receipts	Simple digital receipts/invoices	Must have
AI Assistant	Business questions and recommendations	Core differentiator
Audit Trail	Hash-chained, immutable log of every record created, edited or deleted	Core differentiator
Growth Engine	Cash-flow forecasts, churn risk, credit readiness and auto-generated playbooks	Core differentiator
Marketing AI	Captions, adverts and promotions	Phase 2
Notifications	Low-stock and debt reminders	Phase 2
Staff	Controlled employee access with role permissions	Phase 2
Payments	Subscription billing	Launch

3. Audit & Compliance Module

ChiiTech's natural advantage is that it already captures every sale, expense, stock movement and customer balance at the source. The audit module turns that same data into something an accountant, bank, investor or tax office can trust, without asking the owner to keep a second set of books.

Capability	What it does	Priority
Audit trail & activity log	Time-stamped, user-stamped record of every create, edit and delete across sales, expenses, stock and customer records. Entries cannot be silently altered.	Core
Financial statement generator	Auto-produces Income Statement, Balance Sheet and Cash Flow summaries from day-to-day records, no manual bookkeeping.	Core
Bank & cash reconciliation	Matches recorded sales/expenses against uploaded bank or POS statements and flags unmatched entries.	Core
Anomaly & risk detection	AI flags duplicate entries, backdated transactions, round-number patterns, and expenses that deviate from normal trend.	Core differentiator
Tax & compliance readiness	Tracks VAT, WHT and other levies; produces FIRS/state-ready summaries an accountant can file from.	Phase 2
Approval workflows	Multi-level sign-off before a large expense, discount or stock write-off is posted, so one person cannot alter the books alone.	Phase 2
Document & evidence vault	Every transaction can carry an attached receipt, invoice or photo, stored against tampering.	Phase 2
Auditor collaboration mode	Read-only, time-boxed access for an external accountant or auditor, plus one-click export of an audit package.	Phase 2
Business health & risk score	A single score on the dashboard summarising bookkeeping consistency, reconciliation status and open red flags.	Phase 2
Version history on records	Every edited record keeps its prior versions and shows exactly who changed what, and when.	Core
Regulatory export	One-click PDF/Excel export formatted for regulators, lenders or grant applications.	Phase 3

Design rule: audit features must stay invisible in daily use. The owner records a sale the same way as before; the audit trail, reconciliation and risk flags happen automatically in the background and only surface when useful.

4. UI / UX Design — dark mode, neon-green identity

Design goal: mobile-first, dark-themed interface that feels precise and trustworthy, like a security or fintech console, while staying simple enough for a non-technical shop owner.

Design tokens

Base background	#0A0F0D — near-black, slight green undertone
Surface / card	#141B18 — raised panels and cards
Accent (neon green)	#39FF88 — primary actions, active states, key numbers
Accent deep	#12A64C — pressed states, chart lines, headers on light pages
Primary text	#F3F6F4 — high-contrast white-green
Muted text	#9AA69F — secondary labels, timestamps
Danger / risk flag	#FF5C5C — anomaly and overdue alerts

Typography & layout

Sans-serif throughout (Inter or Space Grotesk). Numbers are the hero: financial figures are set larger and bolder than surrounding labels, always in neon green when positive/neutral and red when a flag is raised. Bottom navigation on mobile, collapsible sidebar on desktop. Generous 12-16px spacing between cards so the dark background reads as calm, not heavy.

Element

Element	Recommended design
Overall look	Dark console aesthetic — flat surfaces, no gradients, one neon accent
Navigation	Bottom nav on mobile (Home, Sales, Products, Audit, AI); sidebar on desktop
Primary action	Neon-green “Record Sale” button, always reachable with one thumb
Charts	Sales trend, profit trend, reconciliation status — neon line on dark grid
Risk / audit flags	Red badge with a short flag label, never buried in a submenu
Feedback	Toast messages; success in neon green, errors in red, both on dark surface
Accessibility	Minimum 4.5:1 contrast kept even on dark surfaces; large tap targets

5. Screen Wireframes

Login / sign-up

CHIITECH

Run your business smarter. Stay audit-ready.

[Email / phone] [Password]

[SIGN IN — neon green button]

New here? Create account

Main dashboard

Good morning, Business Owner

Today: ₦185,000 sales

[Sales ₦185k] [Profit ₦52k] [Expenses ₦31k] [Low stock 7]

Sales trend — neon line chart on dark grid

Business health score: 92/100 • 1 reconciliation flag

[+ RECORD SALE — large neon button]

Home Sales Products Audit AI

AI business assistant

AI BUSINESS ASSISTANT

You: Why did profit fall this week?

AI: Sales fell 8% while expenses rose 14%.

Transport cost was the biggest driver.

Suggested action: reduce delivery costs,
promote your top-selling items.

Ask anything about your business...

What's new in this version

This revision adds the advanced audit intelligence and business growth models that now ship in the first working frontend build, alongside the frontend itself.

New: advanced audit intelligence

Capability	What it adds
Hash-chained audit log	Every log entry embeds the previous entry's hash, so an edited or deleted history entry visibly breaks the chain instead of disappearing quietly.
Benford's Law check	Compares the leading-digit distribution of sale amounts to the pattern genuine transactions naturally follow, surfacing invented or rounded figures.
Expense anomaly rule	Flags any expense 3x+ above the running average, or suspiciously round, before it is trusted as a clean record.
Audit confidence score	A separate score from the general business health score, shown on the Audit page, reflecting flags and log integrity specifically.
Exportable audit package	One click produces a JSON package (ledger, audit log, health score) ready to hand to an accountant, bank or investor.

New: business growth engine

Capability	What it adds
Cash-flow forecast	Projects the next 30 days of cash from the last 7 days of recorded sales, shown as a weekly bar chart.
Best-seller detection	Surfaces the top product by revenue and by units sold, computed from actual sales history.
Customer lifetime value & churn risk	Tracks total spend per customer and flags anyone who has gone quiet for 2+ weeks.
Loan / credit readiness score	A bankable / building / not-yet read on whether the business's own records are currently strong enough to support a loan or grant application.
Auto-generated growth playbooks	Restocking, bundling, win-back and collections suggestions generated from the business's own data rather than generic tips.

New: working frontend

A full click-through frontend now exists (HTML/CSS/JS, dark + neon-green theme) covering Dashboard, Sales, Products, Customers, Expenses, Audit trail, Growth engine and AI assistant. It runs standalone against local storage today; the data layer is isolated into a single file so it can be swapped for the FastAPI/PostgreSQL backend without touching the rest of the app.

What's new — version 3 (multi-company & professional audit)

This revision turns ChiiTech from a single-shop prototype into a multi-tenant platform: any business can register itself, admins control exactly what their workers can see, and one platform owner account can see every company on ChiiTech. The audit and growth modules were also upgraded toward professional, textbook-standard practice.

New: company structure & access control

Capability	What it adds
Two-tier login	A single login screen splits into “Log in as Admin” and “Log in as Worker”, plus a self-serve “Register your company” flow for new businesses.
Company codes	Every registered company gets a unique code (e.g. CHI-1234) that its admin shares with workers so they can sign in under the correct business.
Departments	Eight assignable departments (Sales, Products & Inventory, Expenses, Customers, Orders, Tax, Audit-view, Data analysis & growth) gate exactly which sections a worker can open.
Team & departments page	Admins add workers, tick which departments they can access, and pause or reactivate accounts — all logged to the audit trail.
Platform (super admin) console	A single reserved account sees every company registered on ChiiTech, with team size, plan and join date, and can open any company in a read-only “view as admin” mode.

New: sections

Section	What it does
Tax analysis	VAT and withholding tax computed automatically from recorded sales/expenses, with configurable rates and a CSV export for an accountant.
Orders (remote/non-visitors)	A separate pipeline for WhatsApp, Instagram, phone and delivery-app orders — customers who never physically visited the shop — with a status tracker (Pending → Processing → Delivered).
Billing & payouts	Shows the ChiiTech subscription plan and prepares (but does not yet move real money for) a payout connection for collecting customer payments — that step needs a licensed provider such as Payscale or Flutterwave.

Upgraded: audit module

Capability	What it adds
Plain-language glossary	Every audit term used in the app (audit trail, hash chain, segregation of duties, materiality, Benford's Law, trial balance, audit opinion, reconciliation) is explained in the app itself.
Audit opinion	A Clean / Qualified / Adverse verdict, the same three-tier language real auditors use, derived from the audit confidence score.
Trial balance check	An internal consistency check on recorded cash in vs. cash out, clearly labelled as a precursor to — not a replacement for — full bank reconciliation.
Segregation-of-duties matrix	Shows, per team member, whether they can record transactions, edit prices, and view the audit trail — flagging when one person holds all three.
Three export formats	Audit reports now open as a browser-viewable report, or download as CSV or JSON — no longer a single file forced open in a text editor.

Upgraded: growth engine & sales

Capability	What it adds
Chart type selector	Bar, line, doughnut, or a CSS-based “3D style” isometric view — switchable per chart, on both the Dashboard and Growth engine.
Automated vs. manual analysis	Admins can switch the Growth engine to “Manual” so a dedicated data analyst can override the cash forecast and credit-readiness figures instead of relying on the automatic calculation.
Four-pillar growth plan	Suggestions are now grouped under Sales, Marketing, Finance and Operations, so growth advice covers the whole business, not just restocking.
Cart-based sales with VAT	Sales now support multiple products per transaction, a discount field, automatic VAT, and a generated invoice number.
Editable prices with attribution	Admins and Sales/Products-department workers can edit a product's price directly; every change is written to the audit trail with who made it.
Custom expense categories & root-cause panel	Anyone logging an expense can add a new category on the fly, and every flagged expense now comes with a specific, category-aware suggested fix.

Fixed: mobile responsiveness

The sidebar is now a slide-in drawer on phones (opened from a hamburger menu), reaching every section — previously only five of the app's twelve sections were reachable on a phone. A layout bug that caused horizontal overflow on narrow screens was also fixed.

What's new — version 4 (charts fixed, security & polish)

Testing on a real device surfaced a serious bug: every chart in the app depended on a third-party charting library loaded from a CDN, and that request was silently blocked on the network being used to test it — so every chart stayed blank with no visible error. This version removes that dependency entirely, and fixes several responsiveness, security and clarity issues found in the same review.

Fixed: charts now always render

Fix	What changed
Dependency-free chart engine	Every chart (bar, stacked bar, line, doughnut, and the Benford's Law bar+line combo) is now drawn directly with the browser's native Canvas API in a new <code>js/charts.js</code> — no CDN, no external library, works fully offline.
Dashboard: 3 chart types + 2 new charts	The sales trend chart gained a bar/line/doughnut switcher; two new charts were added — a stacked Revenue vs. Cost of goods chart and a Sales by payment method doughnut.
Growth engine charts fixed	Bar, line and doughnut views (previously blank) now render correctly alongside the existing 3D-style CSS view.
Icon fallback	Icons no longer depend on a font CDN either — they render as lightweight CSS glyphs so the interface never looks broken on a restricted network.

New: platform revenue for the super admin

The Super Admin console now has a “Platform revenue” panel showing monthly recurring revenue across every company, average revenue per company, and a chart of revenue by plan — a direct answer to “how do I make money from this as the platform owner”, separate from each company's own Billing page (which is about that business collecting payments from its own customers).

Fixed: responsiveness & usability

Fix	What changed
Company registration visibility	A highlighted callout now points new users to “Create your company account” directly from the sign-in screen, instead of a small, easy-to-miss subtab.
Zoomed-out / wide-screen centring	Page content now centres itself in the space beside the sidebar instead of hugging the left edge with empty space on the right.
“Getting started” checklist	A dynamic checklist on the Dashboard (add a product, record a sale, invite a worker, log an expense) gives first-time users something concrete to do at a glance, and disappears once complete.

Fix	What changed
Automation toggle broadened	The Growth engine's Automated/Manual switch is no longer admin-only — any worker granted “Data analysis & growth” access can use it too.

Fixed: security

Every place a user-entered name, note, or category is displayed back on screen (tables, the audit log, AI chat) is now passed through an escaping function before being inserted into the page, closing a stored cross-site-scripting gap. This is honestly scoped: passwords are still stored in plain text in the browser and department permissions are still enforced client-side rather than by a server — both require the real backend from the roadmap to fix properly.

What's new — version 5 (critical fixes from real-device testing)

Testing on the person's own device surfaced the most serious bug found in this project so far: every newly registered company was silently seeded with the same sample products, customers and orders as the demo account, making it look like data was leaking between companies. This version fixes that at the root, plus a long list of usability issues from the same round of feedback.

Issue reported	Fix
New companies started with demo data	Fixed at the source: real registrations now get a genuinely empty business record. This was the root cause of several reported bugs, not a separate issue for each.
Chart legends overlapping captions	Charts now reserve their own space for a legend instead of it spilling onto the text below.
Full-width impersonation banner	Replaced with a small corner badge.
Billing page showed fake "Paid" invoices	Now shows an honest "no invoices yet" state, and names Paystack + bank transfer specifically as the two payout methods.
Sales only supported a manual discount field	Now supports entering the final sold price directly, with the discount calculated and shown automatically.
No way to find a forgotten password or company code	Added a self-service flow: admins can reset their own password, anyone can look up their company code by email, and admins can generate a new temporary password for a worker.
No "back" navigation	Added a working Back button that returns to the previous section.
No About/Contact page	Added, reachable from the login screen — explains the problem ChiiTech solves and who it's for.
Growth engine showed only totals	Added a multi-product performance chart (one line per product) with a selectable timeframe from seconds up to 5 years.
Audit page had no way to document how the business should run	Added a Standard Operating Procedures section — write down procurement, evaluation and reporting practices so there's a standard to audit against, with review-frequency and overdue tracking.
Long tables had no way to collapse	Every long list now has a "Show more / Show less" control, implemented at runtime in JavaScript specifically so it can't corrupt the page's HTML the way a text-editing approach did during this round.
Plain black background	Replaced with a subtle, slowly-drifting grid-and-glow texture in the brand colours, on every page.

One fix introduced its own bug mid-build — a script used to add the collapsible-list feature briefly corrupted the page's HTML structure. It was caught by testing before delivery, root-caused, and replaced with a safer approach (wrapping tables at runtime with JavaScript rather than editing raw HTML text). Noted here in the interest of an accurate record, not just the parts that went smoothly.

What's new — version 6 (performance fix, security & legal pass)

A reported lag while using the app traced back to a continuous background animation introduced for visual polish — fixed by making it static. The same round worked through two handwritten checklists: one for not getting hacked, one for not getting sued.

Fixed: performance

The decorative background's continuous 34-second CSS animation forced the browser to repaint a full-viewport gradient layer constantly — the direct cause of the reported lag, including the delay between typing and seeing it appear. It is now fully static. The login page also previously had an opaque background hiding this layer entirely; both pages now look consistent.

Security checklist — addressed

Item	What was done
Password hashing	Passwords are now hashed (SHA-256 + per-user random salt) before storage, replacing plain text. Honestly caveated: this happens client-side, so it raises the bar significantly but isn't equivalent to server-side auth.
Rate limiting (login)	5 failed sign-in attempts locks that email out for 30 seconds, client-side.
Input sanitization	Reconfirmed: all user-entered text is escaped before being displayed back (carried over from the previous round).

Not applicable to this build, and explained as such rather than faked: protecting admin routes, securing API endpoints, CORS settings, securing database access, server-side rate limiting, SQL/NoSQL injection protection, checking exposed files, and checking git history for secrets. These are all things that only exist once there is a real server and database — this build is a static, client-only frontend with neither. They become real work items the moment the FastAPI/PostgreSQL backend from this PRD gets built.

Legal checklist — addressed

Added four legal pages — Privacy Policy, Terms & Conditions, Cookie & Data Policy, and Refund Policy — reachable from the login and About screens. These are templates, not legal advice, and say so on the page; a Nigerian lawyer should review them, and real business details/dates need filling in before launch. Also addressed: keyboard-accessible tooltips, aria-labels on icon-only buttons, and a keyboard-operable mobile menu.

6. User Flow

Create account → Set up business → Add products → Record first sale → Dashboard calculates sales/profit/stock → Ask AI → Receive recommendation → Audit trail logs everything in the background → Return daily.

UX rule: minimize typing. Use product search, quick-add buttons, recent products, numeric inputs and sensible defaults. Audit logging must never add an extra step for the owner.

7. Pricing Model

Plan	Price	Target
Free	₦0	Trial / very small businesses
Founding	₦2,500/month	First 50-100 customers
Pro	₦5,000/month	Main plan
Business	₦10,000/month	Growing businesses + staff + audit trail
Enterprise	₦25,000+/month	Multiple branches, full audit & compliance suite

Gross subscription revenue examples: 100 customers × ₦5,000 = ₦500,000/month; 500 = ₦2.5m/month; 1,000 = ₦5m/month. These are revenue figures, not profit.

8. Technical Architecture

Frontend: HTML/CSS/JavaScript for the MVP; React/Next.js as the product scales.

Backend: Python FastAPI or Django.

Database: PostgreSQL.

Security: password hashing, secure sessions/JWT, role-based access, validation, rate limiting, backups, HTTPS, and immutable audit logs.

AI: LLM API connected to controlled business-data and audit tools.

Payments: Nigerian payment provider for subscriptions.

9. Database Blueprint

Table	Important fields
users	id, name, email/phone, password_hash, role
businesses	id, owner_id, name, category, currency
products	id, business_id, name, sku, cost_price, selling_price, stock_qty, reorder_level
sales	id, business_id, customer_id, total, payment_method, created_at
sale_items	id, sale_id, product_id, quantity, unit_price, cost
expenses	id, business_id, category, amount, note, created_at
customers	id, business_id, name, phone, balance_due
payments	id, business_id, amount, provider, status, reference
subscriptions	id, business_id, plan, status, renewal_date
ai_logs	id, business_id, prompt_type, created_at, usage
audit_logs	id, business_id, user_id, table_name, record_id, action, old_value, new_value, hash, prev_hash, created_at
reconciliations	id, business_id, period, source, matched_count, flagged_count, status
growth_snapshots	id, business_id, period, cashflow_forecast, bestseller_id, credit_readiness, created_at
companies	id, name, owner_email, plan, code, created_at
team_members	id, company_id, email, name, role (company_admin/worker), departments[], active
orders	id, business_id, channel, customer_name, items, total, status, placed_at
tax_settings	business_id, vat_rate, wht_rate, wht_threshold

10. Build Roadmap

Phase 1 — Validate: Interview 20-30 business owners and choose one niche.

Phase 2 — MVP: Login, business setup, products, sales, expenses, customers, dashboard, reports and core audit trail.

Phase 3 — AI: Business Q&A, summaries, stock recommendations, anomaly detection and marketing generation.

Phase 4 — Audit & launch: Reconciliation, financial statements, approval workflows, payments, onboarding, backups, security and support.

Phase 5 — Growth: Staff accounts, auditor collaboration mode, notifications, mobile app, multiple branches and integrations.

11. Best Starting Niches

Fashion sellers/boutiques • Electronics shops • Provisions stores • Salons • Small restaurants. Start with one niche and expand only after proving the workflow.

12. Launch Checklist

- Interview 20-30 owners
- Select first niche
- Finalize MVP
- Design clickable prototype (dark / neon-green theme)
- Build backend/database with audit_logs from day one
- Build mobile-first frontend
- Add authentication/security
- Add AI assistant
- Add audit trail, reconciliation and risk flags
- Add subscription payment
- Test with 5-10 real businesses
- Launch founding plan at ₦2,500/month
- Move toward ₦5,000/month after validation

BOTTOM LINE

Build the smallest useful business manager first. Get real businesses using it, prove they will pay, then layer in audit-grade trust. The AI layer turns business records into clear decisions; the audit layer makes sure those records can be trusted by anyone who needs to check them.